

4.5 Creative task 1 | CMU CS Academy

```
1 app.background = 'black'
2
3
4 ##### Press space to fire at the boss
5 ##### Press left and right to move around and dodge projectiles
6
7 ##### When in phase 2, time your shots to bypass the shields
8
9
10
11
12 ### Player and bolt
13 player = Polygon(180, 350, 185, 370, 215, 370, 220, 350, 210, 350, 210, 360, 205, 360, 205, 345, 195, 345, 195, 360, 190, 360, 190, 350, fill='lime')
14
15 bolt = Rect(195, 320, 10, 20, fill=gradient('black', 'deeppink', start='bottom'), visible=False)
16 bolt.ready=True
17
18 ### Boss
19 #bossbar
20 Rect(80, 15, 240, 20, fill='darkgray', border='gray', borderWidth=5)
21 bossbar = Rect(85, 20, 230, 10, fill='red')
22 #boss shield
23 bossshield = RegularPolygon(200, 120, 40, 5, border='lightblue', borderWidth=5)
24 bossshield.growing=True
25 #boss
26 boss = RegularPolygon(200, 120, 50, 5, fill='magenta', borderWidth=35, border='purple')
27 boss.health=230
28 #hazard
29 hazard = RegularPolygon(200, -300, 10, 3, fill='magenta', border='darkmagenta')
30
31 def gameOverScreen(text,color):
32     Rect(0, 0, 400, 400, fill=color)
33     Label(text, 200, 150, size=70, fill=gradient('black', color, start='top'))
34     app.stop()
35
36
37 def onKeyPress(key):
38     if(key == 'space'):
39         if(bolt.ready == True):
40             bolt.centerX=player.centerX
41             bolt.top=320
42             bolt.visible=True
43             bolt.ready=False
44         if(key == 'left'):
45             if(player.left > 10):
46                 player.centerX-=10
47         if(key == 'right'):
48             if(player.right < 390):
49                 player.centerX+=10
50
```

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51 def onStep():
52     boss.fill='magenta'
53     bolt.centerY-=7
54     if(bolt.bottom <= 0):
55         bolt.ready=True
56     if(boss.health > 0):
57         bossbar.width=boss.health
58     else:
59         bossbar.visible=False
60         gameOverScreen('You Win!', 'green')
61     boss.rotateAngle+=3
62     if(boss.contains(bolt.centerX, bolt.centerY)):
63         if(bolt.ready == False):
64             boss.health-=1
65             boss.fill='white'
66     if(boss.health < 70):
67         if(bossshield.radius < 10):
68             bossshield.growing=True
69         if(bossshield.radius > 80):
70             bossshield.growing=False
71         if(bossshield.growing == True):
72             bossshield.radius+=2
73         else:
74             bossshield.radius-=2
75         if(bossshield.contains(bolt.centerX, bolt.centerY)):
76             bolt.visible=False
77             bolt.ready=True
78     bossshield.rotateAngle-=3
79     bossshield.centerX=boss.centerX
80     bossshield.centerY=boss.centerY
81
82     hazard.centerY+=5
83     hazard.rotateAngle+=2
84     if(hazard.top > 400):
85         hazard.centerX=player.centerX
86         hazard.bottom=0
87     if(player.contains(hazard.centerX, hazard.centerY)):
88         gameOverScreen('You Lose!', 'red')

```